

Research Article

AI in the South Brunswick Classroom: A Student-Centered Analysis

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Abstract

Examining Artificial Intelligence (A.I.) Impact and Perceptions on Education: A Study Focused on South Brunswick High School In order to get some insight, I decided to do a survey in which 668 students in the class of 2026 were surveyed resulting in 400 responses. This survey was constituted by 10 multiple-choice questions, which were administered on Google Forms, covering topics related to AI's impact on learning outcomes, familiarity, and ethics. In order to maximize participation, I first worked with humanities and English composition teachers to spread the survey in their classrooms. The results using this method were limited. So I went up to students during Next, and also organized outreach for outside school with my mom. From September 2023 until April 2024, responses were digitally tracked during the data collection stage. The methodology provided a diverse range of perspectives from a high percentage of the senior class, showcasing students' vastly positive views on AI. Participants focused on AI's promise to improve accessibility, drive personalization and help with real-time feedback. However, the findings also highlighted significant issues, including AI's potential to address questions of inclusivity, ethnic diversity and cultural differences. The participants raised ethical issues including both data privacy and algorithmic bias. In this way, combinatorial approach to data compilation did justice to the complexity and duality of AI in education —highlighting both the transformative potential of this technology while tempering these inherent advantages with the potential negative ramifications of its implementation and use if ethical oversight and inclusion is not part of the development conversation. Educational institutions can utilize the advantages of AI applications while minimizing the risks by implementing ethical supervision and acknowledgement of cultural diversity. These insights are valuable for policymakers, educators, and researchers interested in leveraging AI's capabilities to build more effective and inclusive learning environments.

Keywords: Artificial Intelligence, Data-Analysis, Education, Highschool, Students

1. INTRODUCTION

I am a junior (11th grader) at South Brunswick High School and I've seen many skeptics of Artificial Intelligence (AI) around. Critics say AI undermines traditional education, sacrificing originality and eating at the underpinnings of academic integrity. But as someone who cares about technology and education, I wanted to understand the nuances of this debate — and so I surveyed 400 students in my class about how AI affects key components in education, like learning outcomes, accessibility, and efficiency. AI is the imitation of human intelligence in machines designed to carry out tasks such as problem-solving, decision-making, and learning from existing databases that it has evolved from. AI has the potential to improve education through personalized learning experiences and innovative platforms that enable personalized learning. For example, AI-enabled assistive technologies can help make education more accessible to students with disabilities by providing accommodations appropriate to an students individual needs. A notable example would be Intelligent Tutoring Systems (ITS); they employ adaptive learning algorithms that personalize instruction and address each student's strengths and weaknesses. Learners can achieve their maximum potential while using these systems due to their personalized pacing, detailed feedback, and improved material retention [1]. Studies also show that AI can enhance engagement and outcomes. To provide a glimpse into the impact of AI on education: Knewton, for example is an AI-based platform which increased test score by 62% for 10,000 students [2]. Furthermore, gamified AI applications are proven to increase students' motivation level that makes the learning process much more engaging and fun. While AI brings numerous benefits to society, criticisms surrounding it tend to dominate discussions, with concerns ranging from data privacy issues to the erosion of critical thinking skills. Although these challenges must not be minimized, they also should not distract from the evidence supporting AI's benefits to the field of education. This nuanced perspective will enable better integration of AI, leading to a more inclusive, efficient, and engaging learning environment.

The contemporary education system is being reshaped by AI technology, and it contributes a lot to the engagement of students in the classroom. In the Magazine of Higher Education Teaching and Learning, Distinct Pedagogical Approaches: Understanding the Transformative Potential of Personalization, illustrates the ability of more personalized pedagogies to change and challenge traditional educational practices [3]. AI creates a dynamic and effective learning environment by adapting content and pacing to the needs of each student. Such personalized instruction allowed students to learn at their own pace and to reach their potential and encourage a greater understanding of the curriculum.

From adaptive learning platforms to Intelligent Tutoring Systems, these advancements display how AI has the potential to transform traditional education into more personalized, engaging, and effective experiences. Additionally, AI in education enhances accessibility, allowing students with disabilities to navigate learning hurdles. This paper seeks to analyze numerous aspects of AI in education, whilst also challenging negative perception and survey results, particularly those that have dominated the narratives. I want to illuminate AI's transformative potential while advocating for its careful implementation in education systems.

2. RESULTS

The survey yielded an immense amount of data, which in turn unfolded a plethora of new findings. The first survey question prompted the respondents to scale their familiarity with AI from “not familiar at all” to “very familiar.” This question was curated to be the first because it establishes a baseline understanding of the respondent's knowledge, allowing for a more nuanced interpretation of their subsequent answers. A sizable portion were familiar with the tool, indicating most respondents have a decent understanding of AI's potential in educational contexts (**Figure 1**). Studying the opinion on artificial intelligence's impact on learning outcomes helps assess students' views on how AI influences their learning outcomes, revealing its perceived effectiveness. The data indicated a positive perception of AI's impact on learning outcomes. Most respondents feel that AI improves outcomes (**Figure 2**). Personalized learning paths are a big selling point for the use of machine learning in schools, and this question provides prolific insight as to how students observe changes in their learning. Perceptions on learning path improvement are mostly positive, indicating a general appreciation for tailored content. Amidst this, a minority of the respondents still have mixed feelings about its impact (**Figure 3**). Opinions on student engagement are shown to be divided, indicating that some see AI as detrimental in the classroom. This provides valuable insight into how fields where AI has not had the magnitude of success that many may have envisioned and calls for growth (**Figure 4**). Immediate feedback is a pivotal aspect of AI, and when students were prompted to share their opinions on said skill, the results skewed positively, indicating that students receive and use AI's instant responses in their everyday lives. Only 6% of respondents stated that they never received any sort of feedback from machine learning software. The rest of the 94% consists of subgroups of 54.25%, where people occasionally received feedback and 39.75% who frequently received feedback. Observational data reveals that educational decision-making has undergone significant changes since the introduction of AI, with impacts that encompass both positive and negative aspects (**Figure 6**). Cost and resource efficiency has drastically increased with the introduction of AI, or at least to some extent. The data reveals that an overwhelming majority of respondents believe that AI's implementation has increased efficiency and therefore reduced expenses (**Figure 7**). Responses to questions regarding accessibility and inclusivity reveal diverse perspectives on how effectively AI in education addresses ethnic diversity at high school. While some respondents acknowledged AI's potential to bridge cultural gaps and provide tailored resources for students from various backgrounds, others expressed concerns about inherent biases in AI algorithms and the lack of culturally representative content (**Figure 8**). The introduction of AI in the classroom has resulted in significant policy making and regulation changes in the school environment. Twenty-five percent of respondents experienced some impact, and the rest of the 75% experienced a significant impact on policy making, highlighting the magnitude of AI's impact on education systems and the necessity for schools to adapt to new technological landscapes.

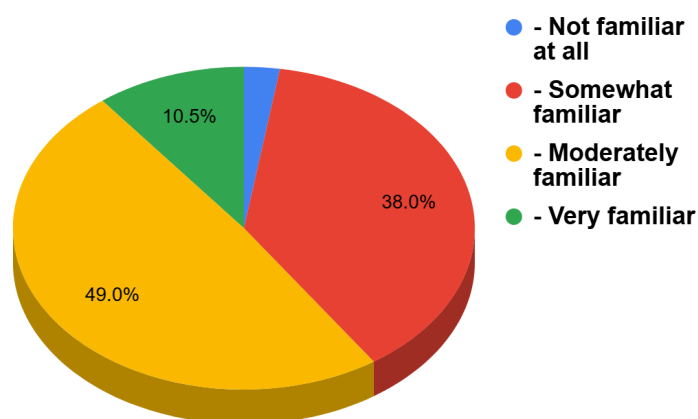


Figure 1. Responses to the question: “What is your current level of familiarity with the use of Artificial Intelligence (AI) in education.?” Pie chart showing the percentage of total respondents that selected any given option. Obtained through survey questionnaire analysis.

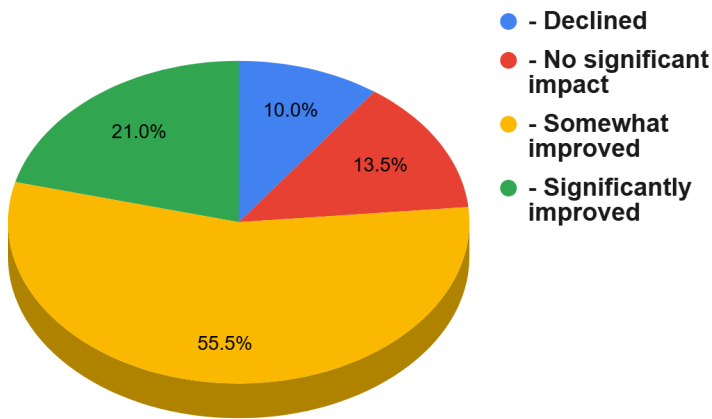


Figure 2. Responses to the question: “In your opinion, how has the introduction of AI impacted learning outcomes in educational settings?” Pie chart showing the percentage of total respondents that selected any given option. Obtained through survey questionnaire analysis.

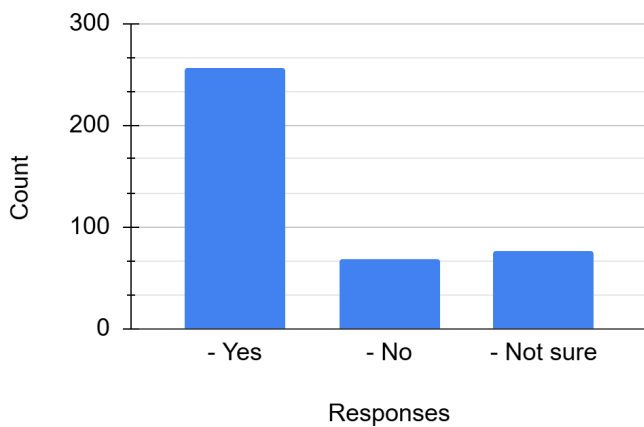


Figure 3. Responses to the question: “Have you experienced or observed personalized learning facilitated by AI technologies in educational environments?” The bar graph shows the number of total respondents that selected any given option. Obtained through survey questionnaire analysis.

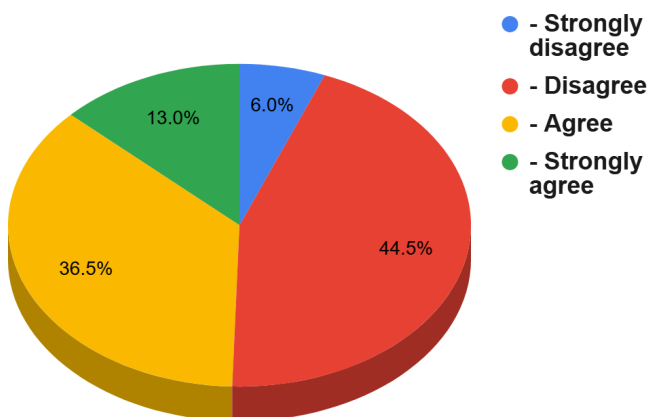


Figure 4. Responses to the question: “Do you believe AI has enhanced student engagement in the learning process?” Pie chart showing the percentage of total respondents that selected any given option. Obtained through survey questionnaire analysis.

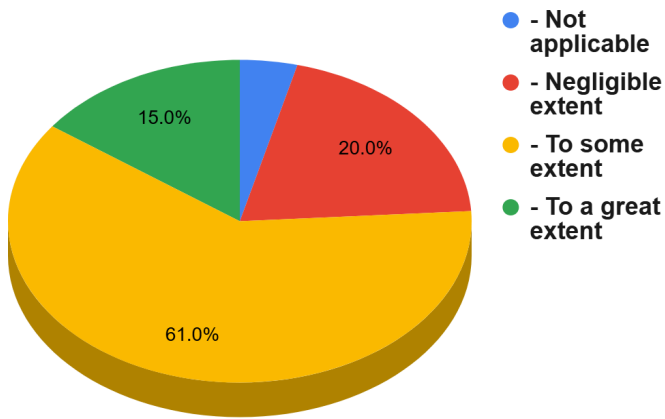


Figure 5. Responses to the question: “To what extent do you think AI-driven data insights have influenced educational decision-making in your experience?” Pie chart showing the percentage of total respondents that selected any given option. Obtained through survey questionnaire analysis.

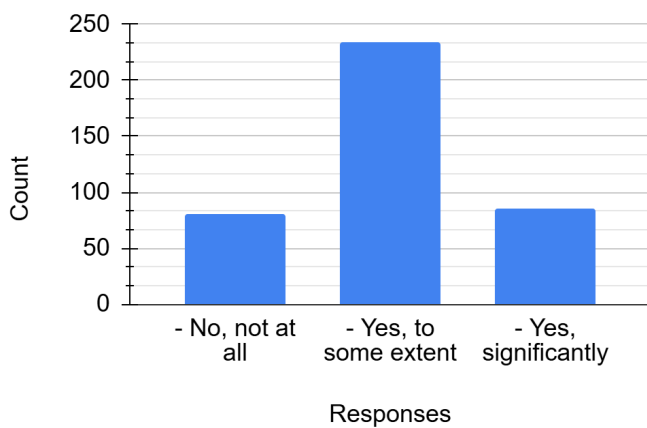


Figure 6. Responses to the question: “In your opinion, has the use of AI in education contributed to cost and resource efficiency within educational institutions?” The bar graph shows the number of total respondents that selected any given option. Obtained through survey questionnaire analysis.

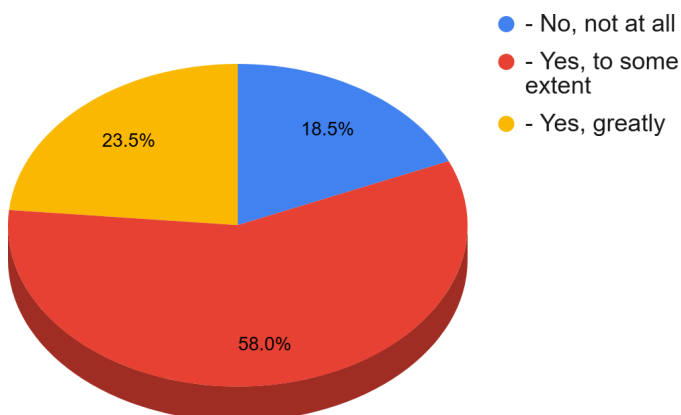


Figure 7. Responses to the question: “From your perspective, has the use of AI in education improved accessibility and inclusivity for diverse student populations?” The bar graph shows the number of total respondents that selected any given option. Obtained through survey questionnaire analysis.

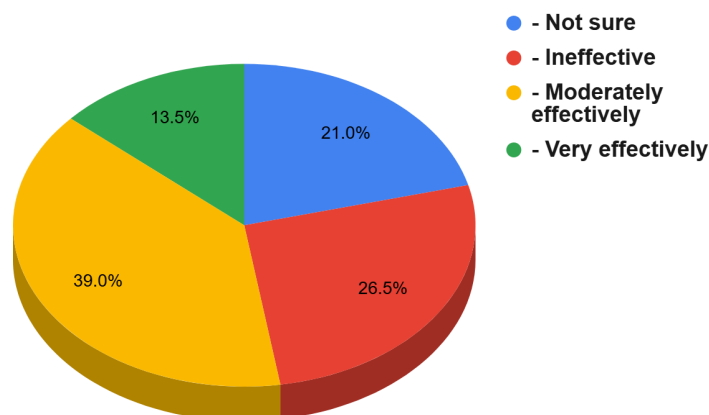


Figure 8. Responses to the question: “How do you think AI in education takes into consideration and addresses ethnic diversity and cultural differences in learning?” Pie chart showing the percentage of total respondents that selected any given option. Obtained through survey questionnaire analysis.

3. DISCUSSION

All statistics referenced in this discussion are available in its entirety in the Figures Section. The figures include all possible answer choices and the prompts to each of the 10 questions.

Student familiarity with AI varied, with almost half (49%) being “Moderately familiar” and only 2.5% having had no familiarity at all. The impact of AI on learning outcomes was also positive for most of the respondents, with 74.5 percent of them noticing or a significant improvement, while only 23.5 percent of the respondents declared no impact or a decline in learning outcomes. Most students had encountered AI personalization in their education, with 64% reporting personal experience with AI-facilitated learning, while the other students either did not know or had never engaged with AI personalization in their learning process. Split down the middle, 49.5% of students disagreed with a positive engagement effect from AI, compared to 50.5% agreeing. The use of AI-powered feedback systems was especially high, with 84% of students receiving immediate feedback through AI tools, and only a small minority never using such systems. Three-quarters of respondents said they believed that AI was influencing decisions made about schools, but they differed on how strongly they believed this was the case. AI received positive results for cost and resource efficiency, over 75% of students believed AI realized enhanced efficiency, of which 21% recorded an impressive reduction in costs. The impact on policy making was universally recognized, and all respondents reported AI influencing some policy or another in their schools, be it integrity policies or BYOD programs. Accessibility + Inclusivity: (81.5%) A total of 81.5% of students agreed on AI increases accessibility and inclusivity in education. The question concerning AI’s effectiveness towards ethnic diversity and cultural variation divided 39% of the respondents claiming that this was moderately effective, 26.5% reporting that this was ineffective, 13.5% thought it was very effective, while 21% revealed that they did not know.

Artificial Intelligence (AI) has increasingly permeated various sectors of society, with education being at the forefront. This rise of AI has also been accompanied by skepticism and concerns, with many teachers and educational critics arguing that it disrupts traditional learning and is overall bad for the educational landscape. As a passionate advocate for technology and its potential in education, I conducted a survey at South Brunswick High School to assess how AI has influenced students’ learning experiences. I will analyze the findings from four hundred responses from my classmates, focusing on understanding student’s familiarity with AI, and its impacts on the pillars of education: learning outcomes, engagement, and accessibility. By examining the collected data, this analysis seeks to find important correlations within the data and address the prevailing negative perceptions that cloud its true potential.

The most striking correlation emerges from the relationship between the familiarity and impact on learning outcome numbers. The data reveals a clear “familiarity advantage,” where those more familiar with AI consistently reported better outcomes across numerous standpoints. For example, 80% of respondents who were “very familiar” with AI, 71.4% who were “moderately familiar” and 66.7% of “somewhat familiar” reported having seen positive learning outcomes through the incorporation of artificial intelligence. All being

said, this relationship is not simply linear. The data also reveals a fascinating nuance: there is a non-linear relationship between AI familiarity and evaluation patterns. As familiarity increases, evaluations become more nuanced and less polarized, suggesting that expertise leads to a deeper understanding of AI's strengths and limitations.

Another complex and revealing interconnection cluster centers around feedback systems. The data shows a three-way relationship between feedback frequency, student engagement, and learning outcomes:

1) Direct Feedback-Outcome Correlation:

- a) 73.3%, with frequent feedback reported improved learning outcomes.
- b) 65.2%, with occasional feedback reported improvements.
- c) Only 33.3%, with no feedback, reported improvements.

2) 81.8% of those with frequent feedback AND positive engagement reported improved outcomes. Only 25% of frequent feedback but negative engagement saw improvements.

This suggests that feedback frequency alone is not sufficient - engagement acts as a crucial mediating factor. The correlation becomes even more interesting when we consider that 86.7% of those receiving frequent feedback responded positively about student engagement, indicating a potential virtuous cycle where feedback enhances engagement, which in turn makes feedback more effective. Examining a correlation between familiarity, data-insights, and cost efficiency develops a unique chain of implementation. This "implementation chain" reveals how various aspects of AI implementation build upon each other:

1) At the extreme highs ("Very Familiar" and extensive data insights):

- a) 100% reported positive cost efficiency.
- b) Seventy-five percent reported significantly improved learning outcomes, and no one believed outcomes had declined.

2) At the middle level ("Moderately Familiar" with moderate insights):

- a) 76.5% reported positive cost efficiency.
- b) 53.8% reported improved outcomes.
- c) 15.4% reported declined outcomes.

3) At the lower positive response level ("Somewhat Familiar" and negligible data driven insights):

- a) 33.3% reported positive cost efficiency.
- b) 15.4% reported significantly improved outcomes.
- c) 15.4% reported declined outcomes.

This chain effect suggests that successful AI implementation requires building from a solid foundation of understanding through effective data utilization to achieve the benefits in schooling systems that so many people expect from it.

One of the strongest socially significant correlations emerges in the relationship between cultural consideration responses and accessibility improvements. This correlation reveals the human dimensions of AI implementation. Among those reporting "greatly" improved accessibility:

- a) 33.3% rated cultural consideration as "Very effective."
- b) 44.4% rated it as "Moderately effective."
- c) 11.1% rated it as "Ineffective."

Conversely, among those reporting no accessibility improvements:

- a) No one rated cultural consideration as "Very effective"
- b) Seventy-five percent rated it as "Ineffective."

This strong correlation suggests that cultural sensitivity and accessibility are not separate concerns but deeply interconnected aspects of effective AI implementation. Furthermore, policy related disconnect can be observed in data insight and policy impact responses. 85.6% of those who reported extensive data insights reported significant policy impacts, however 71.4% of those reporting negligible data insights also reported significant policy impacts. This suggests that policy changes may be driven by factors beyond empirical evidence and raises questions about the alignment between practice and policy for the implementation of AI in education.

This interconnected web of correlations reveals that successful AI implementation in education is not about optimizing single factors but understanding and nurturing the relationships between various aspects that it impacts. More importantly, this data reminds us that AI implementation in education is not just a technical challenge, but also a human one, and requires careful attention to both qualitative human factors and quantitative outcomes. Success lies not in maximizing any single metric but understanding and optimizing the complex relationships between several aspects of the issues present.

4. LITERATURE REVIEW

AI is the revolutionary force in education, transforming the ways in which students learn, collaborate, and access resources. According to research, while it holds a promise for solving some traditional educational issues, ethical and equity concerns arise. Great prospects but also unique challenges are associated with the use of AI in the classroom [4]. This literature review identifies and evaluates contemporary evidence regarding the impacts of AI on personalized learning, student engagement, feedback loops, accessibility, cultural responsiveness, and policy frameworks. Probably the most publicized application of AI in education is its touted ability to provide learners with personalized learning experiences.

Active platforms and Intelligent Tutoring Systems can offer tailored content and assessment to meet individual student needs, therefore allowing for mastery based learning to take place [5]. Evidence suggests that technologies like Knewton and Carnegie Learning realize tremendous gains in improvement when they customize learning material and pace to precisely what the student can do [6]. On the contrary, Kulik and Fletcher (2016) view that these systems promote self-learning and allow students to learn at their own speed [7]. However, it remains inaccessible because of concerns about equity of access to technology across the socio-economic groups of people as noted by VanLehn, 2011 [8]. These results also align with the recent work done at South Brunswick High School, where in student surveys it was found that students generally view AI as a support that is helpful in personal matters and also recognize that access and resource availability differ by demographic characteristics. More detailed information about the specific problems students are experiencing might help in highlighting potential solutions. This is especially true for gamification and interactive learning platforms, which have been successful in increasing student motivation through the use of AI tools.

By using real-time interaction and feedback, these can foster motivation and participation [9]. Hew et al. (2019) that Gamified form of Ai Tools can play positive role on classroom dynamics [10]. However, Kay et al. (2013) also note that the novelty value of such tools may result in short rather than sustained engagement, and their long-term efficacy requires evaluation via longitudinal studies [11]. These observations track with what people found at South Brunswick High School, where students said AI tools can be motivating, even as they raise concerns about their long-term appeal and efficacy in helping maintain focus. This discussion could be strengthened by providing concrete details on some of these tools and how they were implemented. Our AI-driven feedback systems are critical in enhancing learning.

Tools that are used to automate, such as essay grading systems and virtual tutors, provide immediate feedback on the performance thus allowing iterative improvement to be made [12]. Such systems are particularly useful in the Science, Technology, Engineering, and Mathematics (STEM) disciplines, which frequently require instruction by steps. Data from the survey at South Brunswick High School confirms this based on how the value of AI support lies in the instant feedback and building a learning process that could be much more iterative and hands-on. At the same time, however, students warned against complete dependence on AI feedback and mentioned that human input alone can provide critical responses sensitive to context. References to specific examples of student use of the feedback systems would further enrich this analysis. AI can do a lot to make education more accessible. Technologies of assistance, like speech-to-text software, screen readers, and real-time translation apps, have become very important in promoting inclusive learning for students with disabilities and non-native speakers [13]. Furthermore, artificial intelligence can abide by differentiated standard to meet the different learning needs or sedentary students as well as students who required special education [14]. Yet deep-seated biases in AI algorithms present a real threat. As Eubanks (2018) points out, this commonly mirrors societal biases in the risk of excluding under-represented groups. These risks should be mitigated with rigorous ethical oversight. This coincides with concerns from South Brunswick students that the tools be culture sensitive and help them deal with various challenges this increasingly diverse pool of students present. Giving a more fine-grained examination to particular accessibility tools and their function in servicing the needs of particular students can better serve to inform that discussion. Introduction of the opportunity and challenges regarding the use of AI in multicultural classrooms: On the contrary, culturally responsive AI tools can enhance the classroom environment and learning outcomes for the students [15].

However, there is a lack of representation of diverse cultures in AI training datasets; this leads to stereotyping and exclusion of marginalized communities [16]. Overcoming such deficiencies in data will require diversification of sources and the creation of AI-driven tools within the framework of local needs. Responses from South Brunswick High School captured that sentiment, with students calling for the development of AI technologies that include a place for the diverse range of students attending the school. This section could have been greatly strengthened within the piece with examples of successful culturally responsive AI tools. Whom these learning policies were actualized and how they were practiced in classroom environments have been impacted by AI also, for example, design of curriculum and its assessment and educational technologies.

First, there is a lack of longitudinal studies on the long-term impacts of AI on student outcomes and engagement. Second, appropriately, deeper consideration is needed regarding the ethical implications of AI in education, particularly relating to data privacy concerns or algorithm bias. Finally, it is an under-researched area as far as the intersection of AI and cultural diversity in education is concerned, and further studies are needed on the global applicability of AI tools. This article takes a closer look in the case of South Brunswick High School, where students and teachers alike have made it a point to stress that stronger research is needed about how AI is implemented and policies are developed. Figuring out ways to fill those gaps might provide a blueprint for future studies. AI has huge promise to change landscapes at schools, improving personalized learning, engagement, feedback, accessibility, and inclusion.

Yet, all these have to be pitted against ethical considerations and cultural and socio-economic inequities. Still, the outcomes at South Brunswick High School underpin both the promise and challenges of integrating AI into education systems. Students say AI offers them an opportunity to reshape their own learning experiences but should go hand in hand with consideration for equity, cultural responsiveness, and ethical oversight. In this regard, equitable and inclusive educational technologies for diverse populations will have to be designed through active collaboration among researchers, educators, and policymakers in concert with the emerging development of AI.

5. MATERIALS AND METHODS

To understand how students are adapting to the use of AI with regards to the enhancement of learning outcomes and the ethical implications of that implementation, this study utilized a quantitative survey methodology to explore familiarity with AI and its effect on learning outcomes among students. We fixed the survey by using the Google Forms software to a poll from the participants. Data were collected from September 2023 and to April 2024. At first, I reached out to both my humanities and English composition teachers, who consented to place the survey in their individual classrooms. But this approach was far less responsive than expected. I then extended the collection process to include one on one approach of students during free periods at school and also outside of school hours with the support of my mother until the response rate reached the desired level. I was also keeping a log to see how many people responded, like a data sage, making sure this was getting us to the destination we had in mind. There were 668 students total in the South Brunswick High School class of 2026, with 400 responding to the survey. These led to a wide variety of perspectives from the students regarding AI and its implications. The survey comprised ten questions, each with four possible answers. These questions were intended to elicit responses about three areas of interest: Student familiarity with AI, perceived effects of AI on learning outcomes and student views on the ethical aspect of AI use in the context of education. The questions selected for both groups were designed to be as clear and comprehensive as possible so they would apply broadly to many aspects of A.I., while remaining approachable by each respondent. After collecting the responses, I started analyzing the data. Using Google Sheets, they characterized and examined patterns in the responses. Descriptive statistics were used to analyze the data, focusing on the most frequent responses and trends in students' understanding of AI and its impacts. This had allowed the general sentiment toward AI, its potential in education, and a concern regarding its ethical use have helped formulate a dynamic perception of AI in education. To maintain the ethical integrity of the study, the following measures were taken: informed consent was collected from all participants with a proper understanding of the survey purpose. Responses were kept anonymous to protect student privacy. Participation in the survey was entirely voluntary, and students were told they could retract at any point without penalty. Interviews were not conducted in this study, although they would have provided more qualitative data on the students' opinions. Initially, the response rate was lower than expected, possibly reducing the representativeness of the sample.

Abbreviations

AI: Artificial Intelligence

ITS: Intelligent Tutoring Systems

STEM: Science, Technology, Engineering, and Mathematics

BYOD: Bring Your Own Device

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Author Contributions

Shreyansh Dubey is the sole author. The author read and approved the final manuscript.

Conflicts of Interest

The author declares no conflicts of interest.

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